23.11.25
তাওহিদ স্যার


gate → mossfet, diode,

* Memory ← Register ← flipflop ← gate.

* Microchip → LSI
  MSI

* chip design company (bangladeshi, bidoshi)

* Sensor → How they work
↓
কিভাবে বানানো হয়? → how to design

* why digital electronics need for CSE student

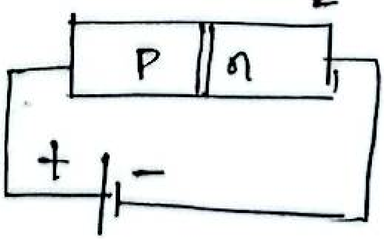
* Soil → specific ~~mete~~parameter এর উপর depend করে sensor তৈরি

* How signal transfer? — analog / digital

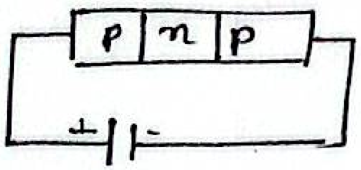
* diode/transistor কিভাবে কাজ করে (work)

* ohms, kcl, kvl,

2

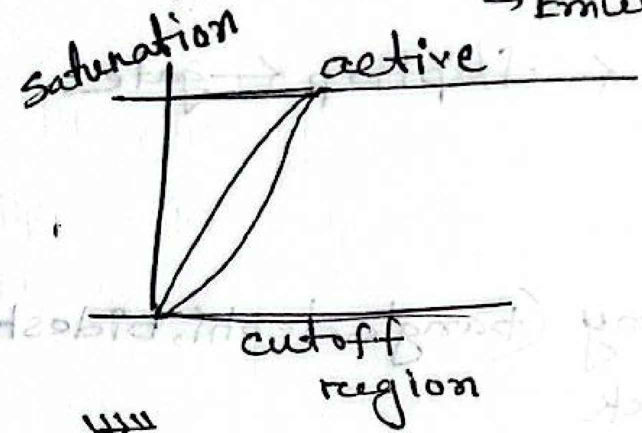


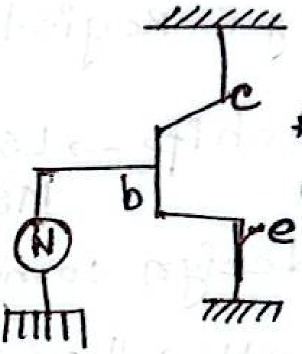
*transistor (use → switch



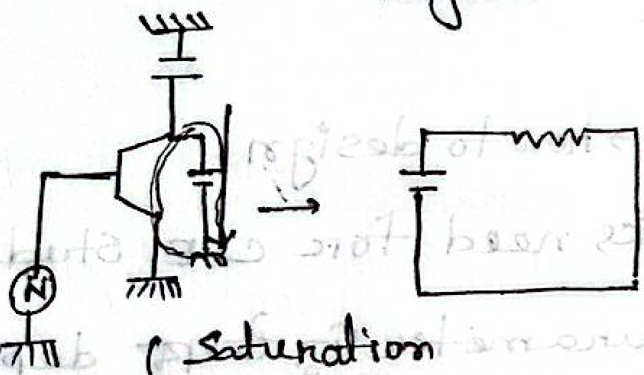

→ Base
→ collector
→ Emulator

- Saturation (+, +)
- cutoff (-, -)
- Active (+, -)




*define which one is base, collector, emitter

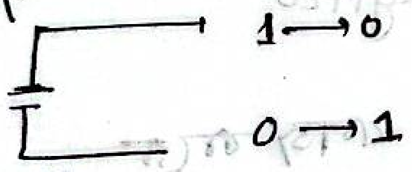



(Saturation both forward)

*NOR, Nand gate কিভাবে তৈরি হবে (Next class).

vcc = 0.2 (0 হওয়া যাবে না)

$V = IR$
$I = V/R$

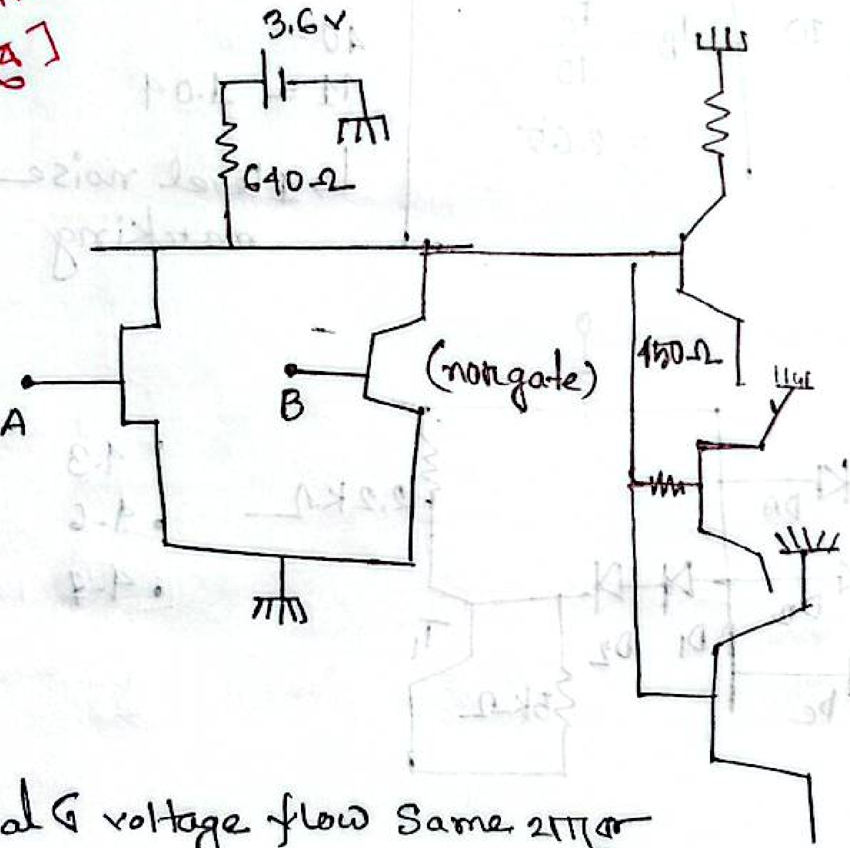


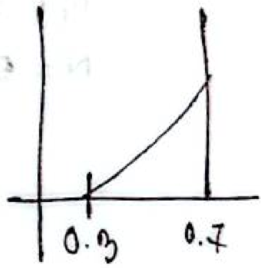

*Technology + Human Scycology

মোবাইল ফোন কিভাবে frequency allocate করে — TDM / CDM

~~ক্লাস~~ 8.12.25
তাওহিদ স্যার

[Question হবে]



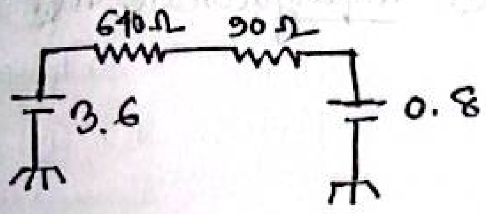




diode characteristic

Parallal এ voltage flow Same থাকে

Series এ current flow Same থাকে

*




$$\frac{3.6-0.8}{640+90} = I$$

$$V = IR$$

$$I_C = \frac{3.6-0.2}{640}$$

$$HFE \cdot I_B \leq I_C$$

$HFE = 10$

$N = 5$

$$I_B = \frac{I_C}{10}$$

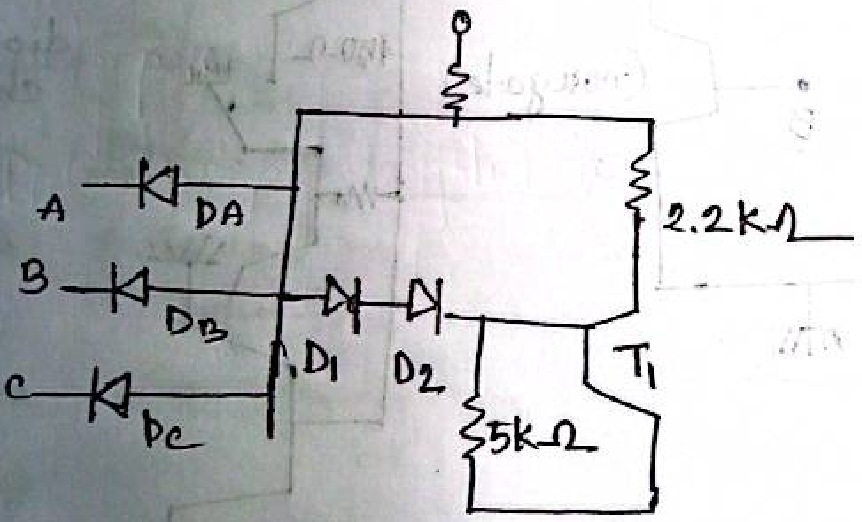
$$= 2.65$$

$40 =$

$41 = 1.04$

↳ 1 level noise marking

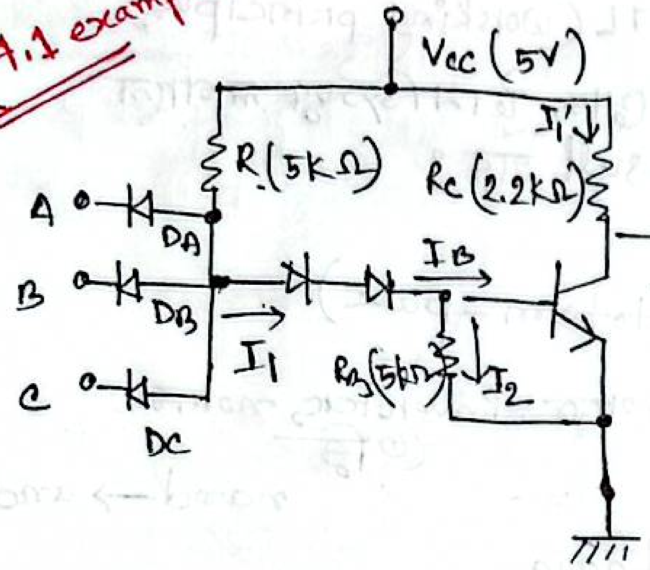
*

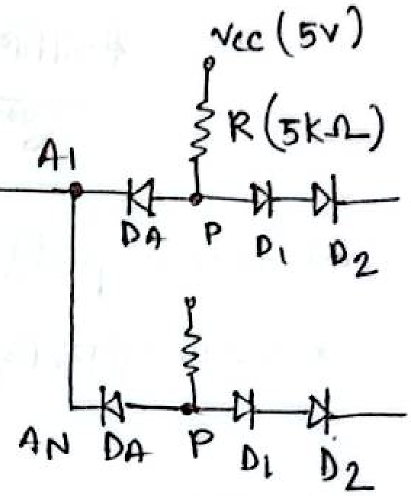



- 4.3
- 4.6
- 4.7

(4.1 example)

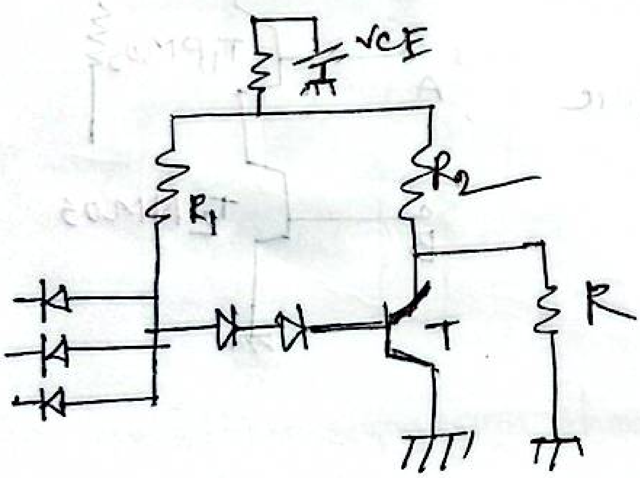
9.12.25



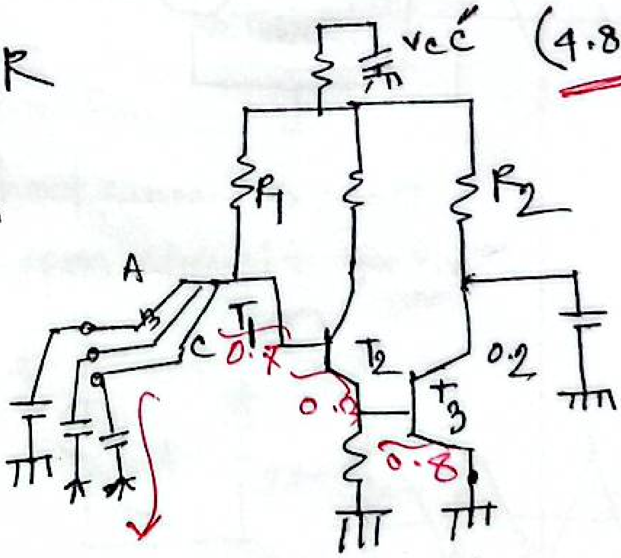




4.01.26

EEE




(4.8.2)




nand gate এর
মতো কাজ করবে

* problem of DTL, TTL (working principle)

*সার্কিটে ডায়োড, ট্রানজিস্টর লাগালে কেমন result হবে ?

* active pull up (totem-pule)

* একাধিক ট্রানজিস্টর থেকে, inverter, nor, or তৈরি

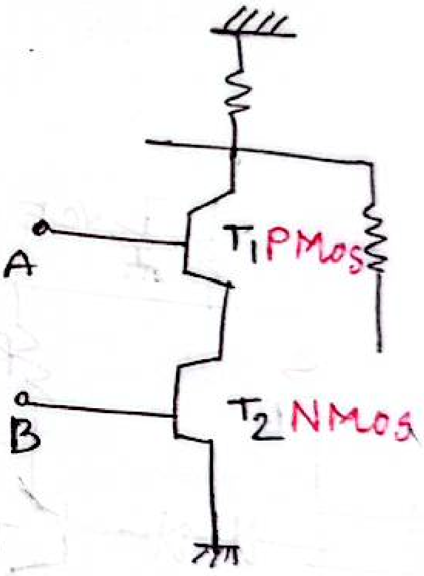
nand → and

## Emitter coupled Logic

CMOS

clamper

operational amplifier



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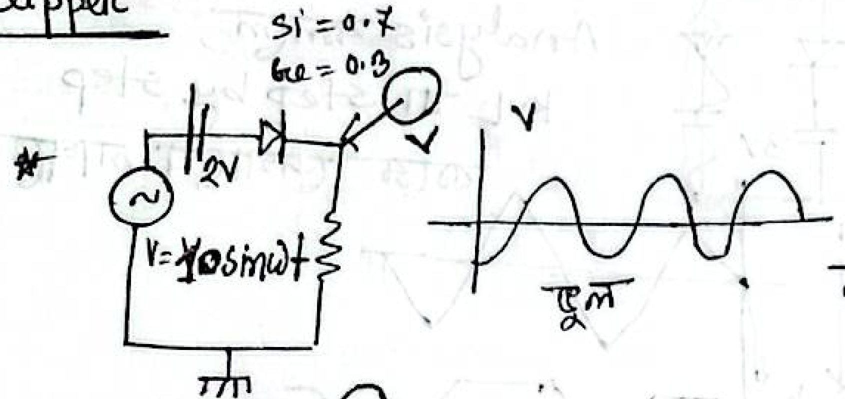
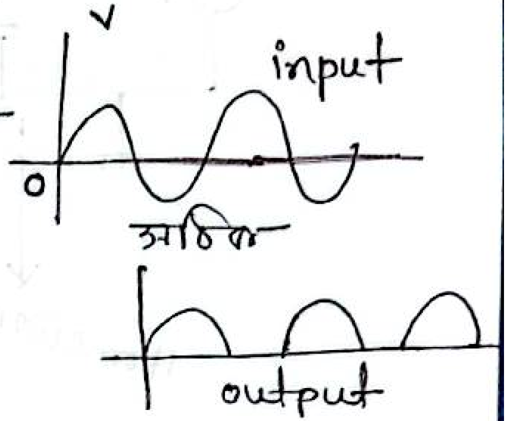
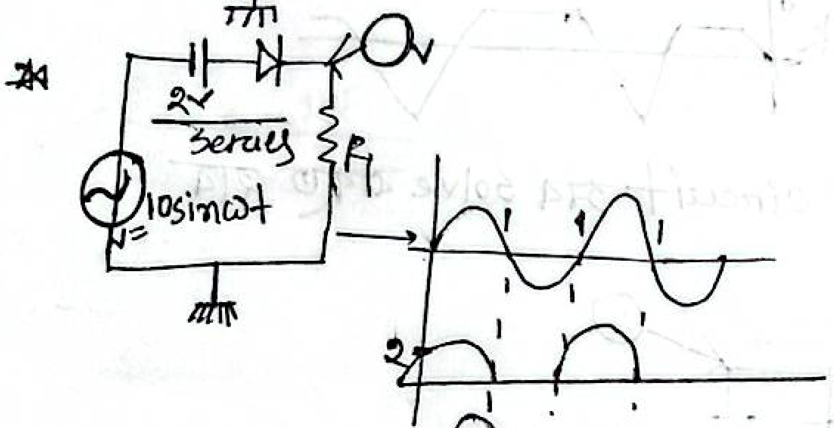
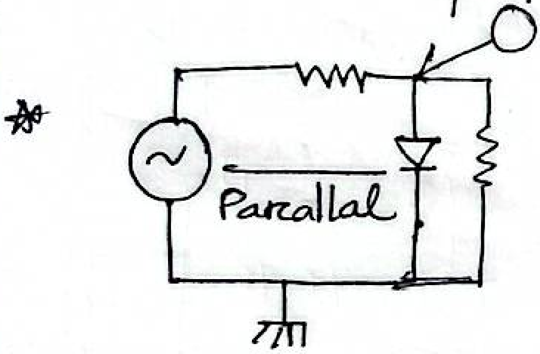
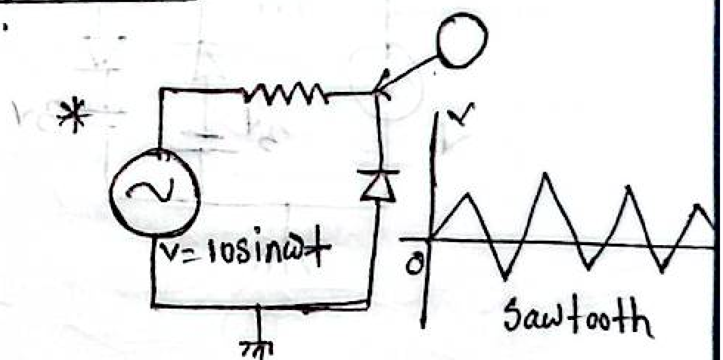
## Clipper



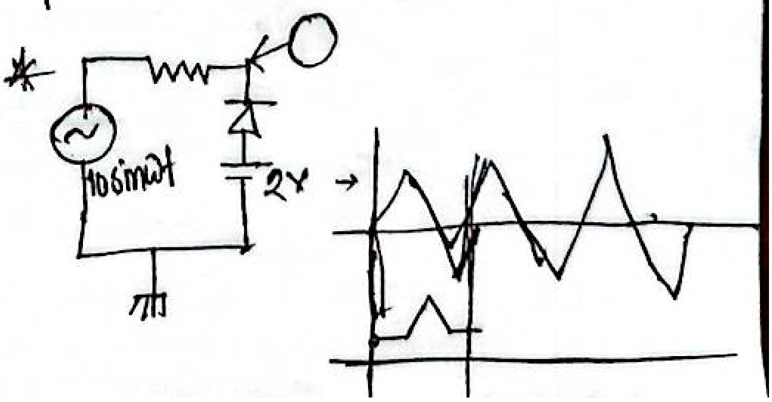



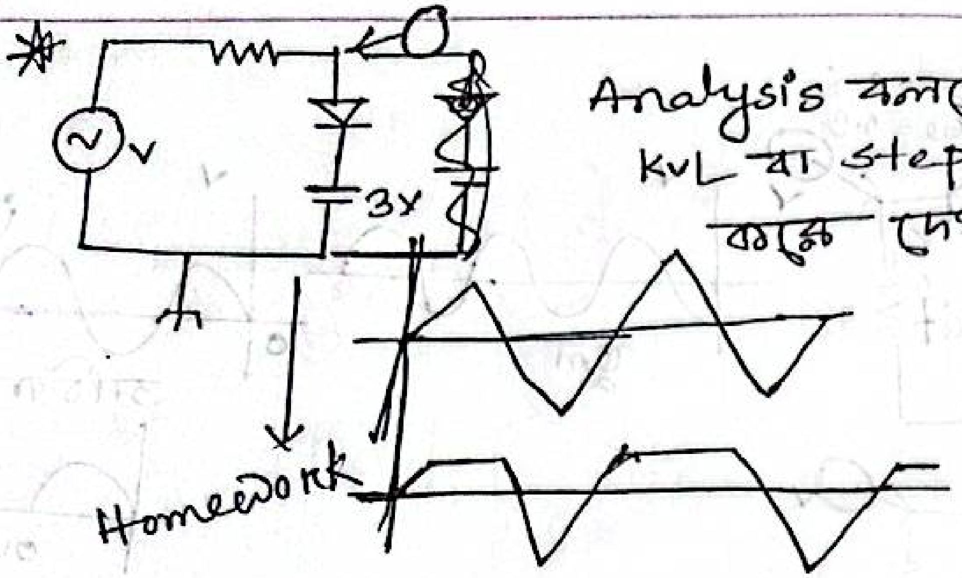




positive অংশে গেলে short circuit, v = 0

negative অংশে গেলে open circuit, v = vin



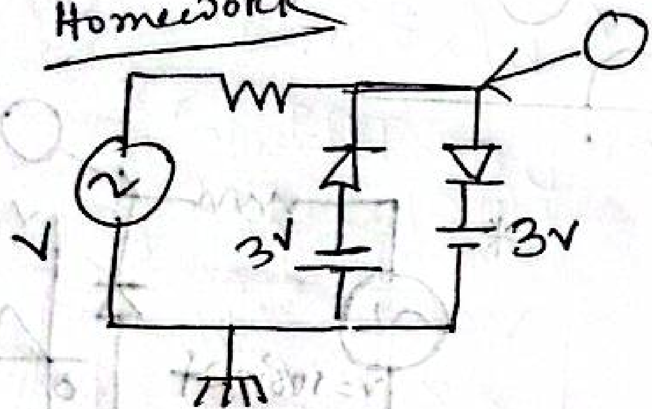
*




Analysis করলে,
KVL বা step by step
করে দেখানো লাগবে

clipper এর circuit সব solve করতে হবে

Homework



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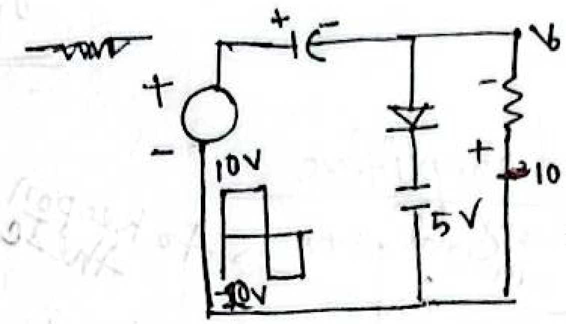
## clamper:

Ex-2.24

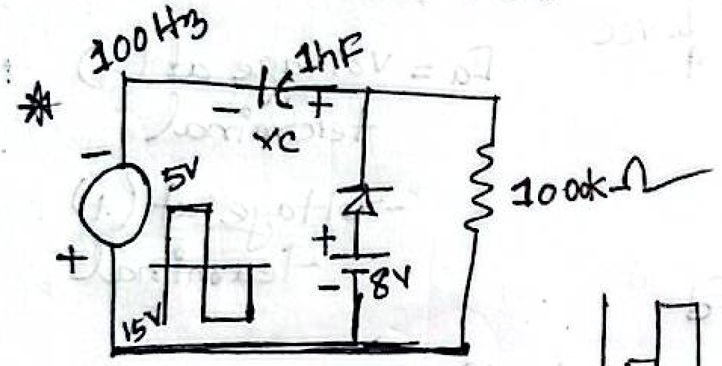


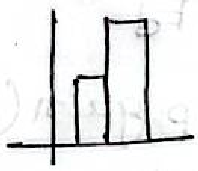

$-10 + V_C - 5V = 0$

$V_C = 15\ V$

$+20V + 15 + V_0 = 0$

$V_0 = 35$



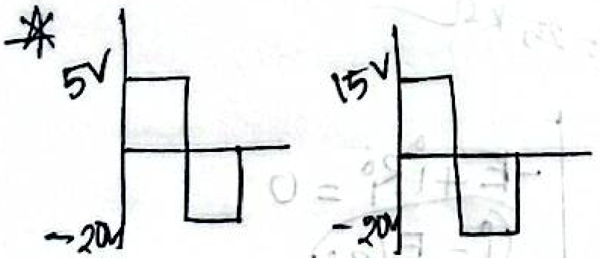
$+15V - V_C + 8V = 0$

$V_C = 23V$

$-5V - 23 + V_0 = 10$

$V_0 = 28V$

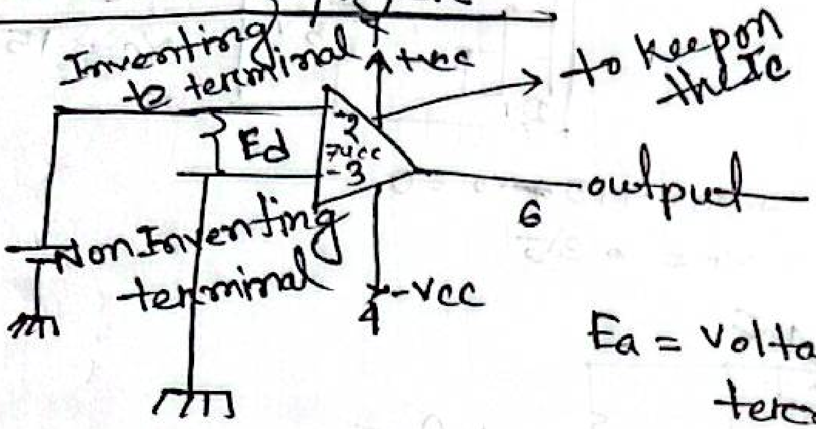
* next class (operational amplifier)




* circuit design করতে হবে ।

07.01.26

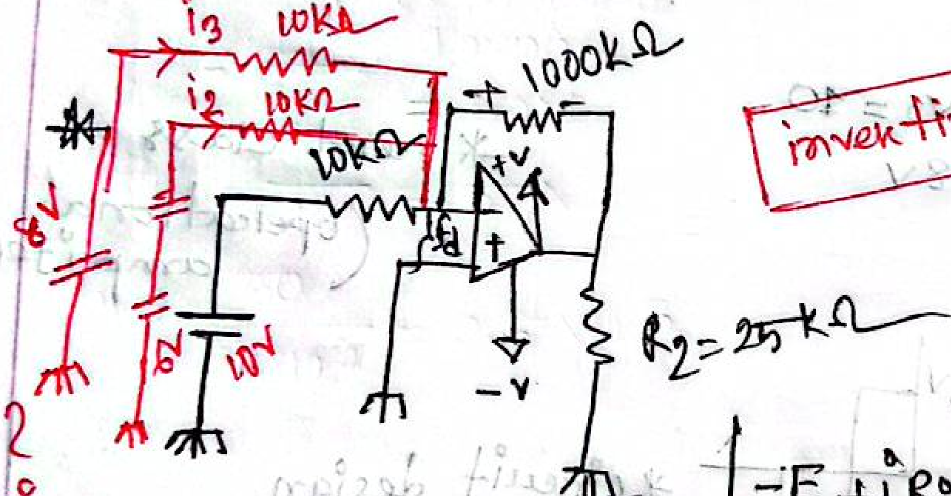
## operational amplifer




Ea = voltage at (-) terminal - voltage at (+) terminal

led (on) $V_o = A_{ol} \times E_d$

Closed Loop gain (Acl)




inverting adder

$V_o = V_{Rf} = -E/R_i \times R_f$

output voltage

$-E + iR_i = 0$

$i = E/R_i$

$V_{Rf} = i \times R_f$

$= E/R_i \times R_f$

$$\boxed{\frac{V_o}{E_i} = -\frac{R_f}{R_i} = A_{cl}}$$

* $E = 10\,V$

input current,

$V_{Rf}$

$A_{cl}$

$V_o$

$i_L$

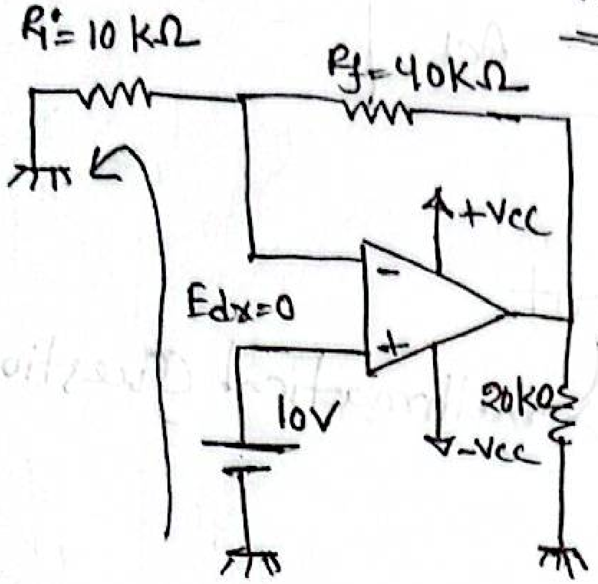
mathmatical Question:

* $R_f = 100\cancel{0}\,k\Omega$

$\therefore V_{Rf} = \frac{1V}{\cancel{10}} \times 100\cancel{0}\,k\Omega$

~~$V_{RF} = 1000\,k\Omega$~~

11.01.26




$i = \frac{E}{R_i}$

$i$ ?

$V_o$ ?

$A_{CL}$ ?

$i_L = ?$

$$V_o = E + V_{Rf}$$
$$= E + iR_f$$
$$= E + \frac{E}{R_i} \times R_f$$
$$= E\left(1 + \frac{R_f}{R_i}\right)$$

$$A_{CL} = \frac{V_o}{E} = 1 + \frac{R_f}{R_i}$$

$$i = \frac{10}{10k\Omega} = \cancel{10A} = 1\,mA$$

$$V_o = E + V_{Rf}$$
$$= 10 + (1 \times 40 \times 10^3)\cancel{k\Omega}$$
$$\cancel{= 50}$$
$$= 50V$$

$$A_{CL} = \frac{50}{10} = 5\,A$$

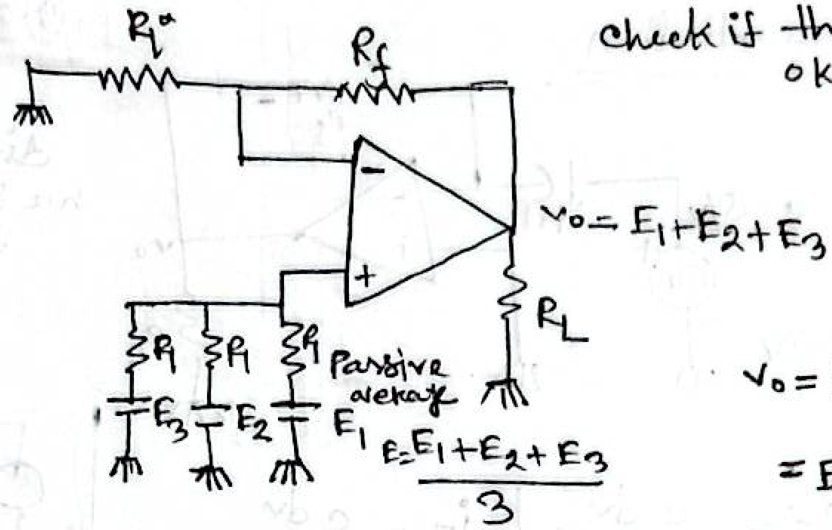
$$i_L = \frac{50}{20 \times 10^3} = 2.5 \times 10^{-3}$$

~~ভোর বুঝিনি~~

~~ভুল দিলে~~

$$A_{CL} = 1 + R_f/R_i$$
$$= 1 + 0 = 1$$

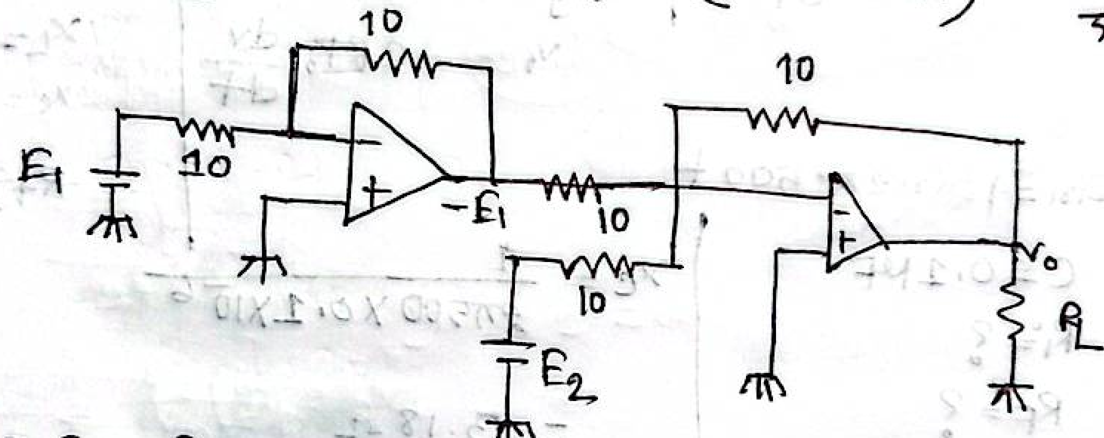
* check if the circuit is ok or not?




$$v_o = E\left(1+\frac{R_f}{R_i}\right)$$
$$= E\left(1+\frac{20}{10}\right)$$

যদি E দিলে
Rf , Ri এর
সাথে 3গুণ
হতো

* $E_1 - E_2$ (Draw the circuit) → (substractor)

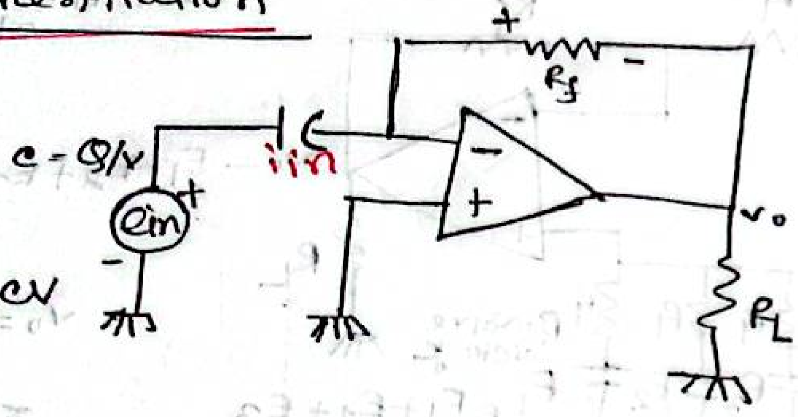



$$v_o = E_1 - \frac{R_f}{R_i}$$
$$= -E_1$$

$$v_o = E_1 + E_2$$
$$= E_1 + (-E_2)$$

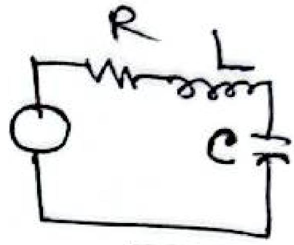
## * Differentiation

13.01.26




difference
hu signal from
500 to 1000

$Q = cv$




$$\frac{dQ}{dt} = c\frac{dv}{dt}$$

$$i_{in} = c\frac{dv}{dt}$$

$$i_F = c\frac{dv}{dt}$$

$$-\frac{v_o}{R_f} = c\frac{dv}{dt}$$

$$v_o = -CR_{F_o}\frac{dv}{dt}$$

$$z = \sqrt{R^2 + (x_L - x_c)^2}$$

$$x_L = \omega L$$

$$x_c = \frac{1}{\omega C}$$

$$R + j(x_L - x_c)$$

$e_{in} = 4\sin 2\pi 500 t$

$C = 0.1\mu F$

$R_1 = ?$

$R_F = ?$

$v_{out} = ?$

$$x_c = \frac{1}{2\pi 500 \times 0.1 \times 10^{-6}}$$

$$= 3.18\,\Omega$$

$$T_1 = \frac{1}{1000}$$

$$T_2 = \frac{1}{1000} \quad [T_2 = 2T_1]$$

$$R_f = \frac{2}{2\pi f c}$$

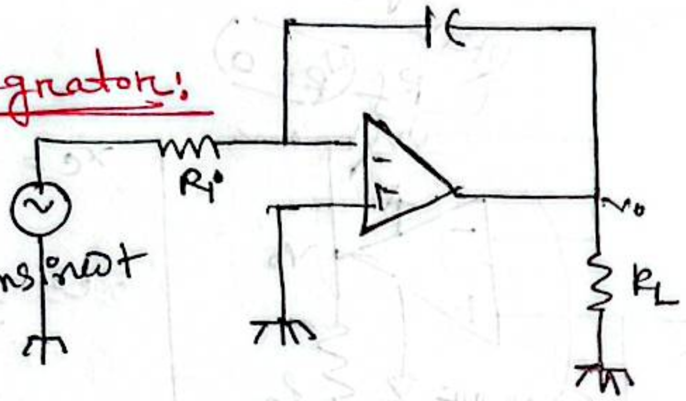
$$R_i = \frac{1}{2\pi f c}$$

$$x_c = \frac{1}{2\pi f c}$$

$v_o =$

$v_o = -R_f C_i \frac{de_{in}}{dt}$ | $T_2 = 2T_1$

## * integrator:




$i = \frac{e_{in}}{R_i}$

$v_c = \frac{Q}{C}$

$= \frac{1}{C} \int i \, dt$

$= \frac{1}{C} \int \frac{e_{in}}{R_i} dt$

$= \frac{1}{RC} \int e_{in} dt$

(next topic filter)

$v_o = -v_c$

$= -1/RC \int e_{in} dt$

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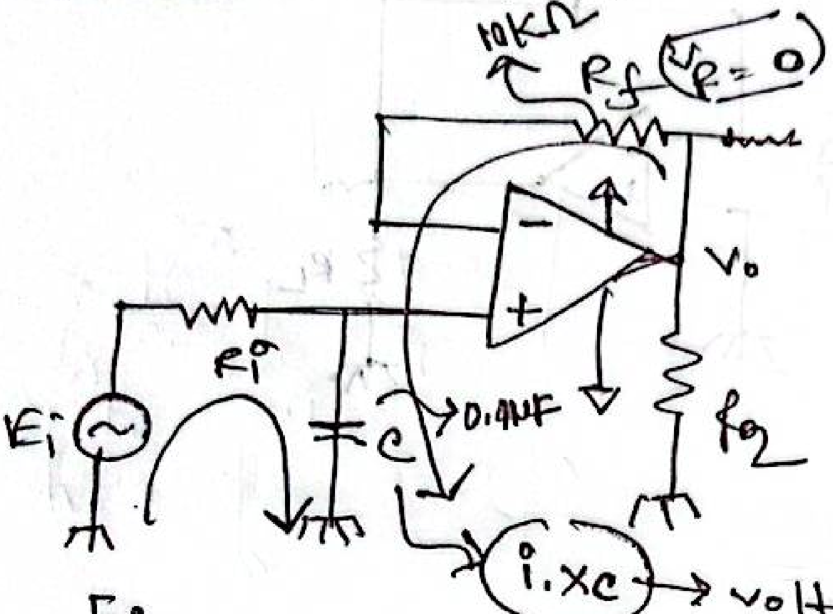
**low pulse (non-inverting)**




$x_c = \frac{1}{j\omega c}$

$x_c = \frac{1}{j2\pi f c}$

$$i = \frac{E_i}{R_i + \frac{1}{j\omega c}}$$

$$v_o = v_c + v_{Rf}$$

$$= v_c = i x_c = \frac{1}{j\omega c} \times \frac{E_i}{R_i + \frac{1}{j\omega c}}$$

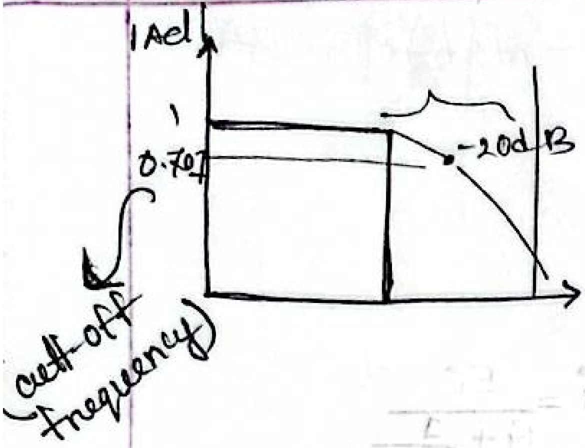
$$\therefore v_o = \frac{E_i}{1 + j\omega RC}$$

$$|A_{cl}| = \frac{v_o}{E_i} = \frac{1}{1 + j\omega RC}$$

$|A_{cl}| = 1$ হলে input = output

$|A_{cl}| = 0$ হলে input ≠ output

$|A_{cl}| < 0$ " input ≠ output




* cuttoff frequency বের কর।

$R = X_C$ ~~[ω = 1]~~ $[j\omega RC = 1]$

$R = \frac{1}{j\omega C}$

$\omega = \frac{1}{RC}$

$$\omega_C = \frac{1}{10 \times 10^3 \times 0.04 \times 10^{-6}}$$

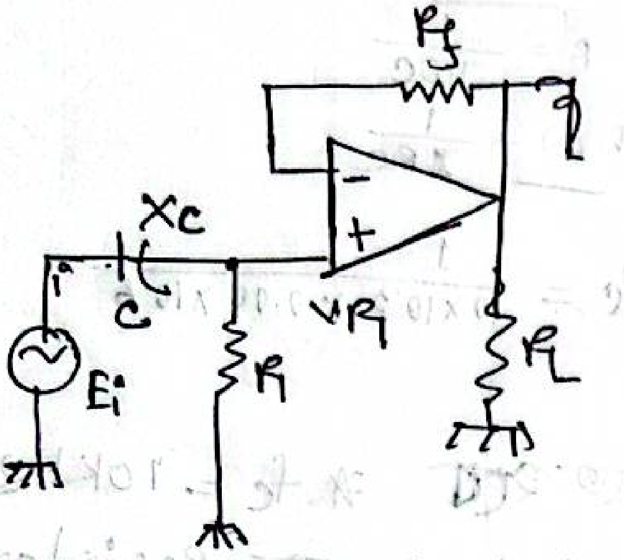
* Circuit design করতে হবে  * $f_c = 10KHz$.

Circuit draw করে Resistance বের কর

inductor দিয়ে Circuit draw করতে হবে

18.01.26

high-pulse




$$i = \frac{E_i}{R_1 + \frac{1}{j\omega C}}$$

$$V_{R_1} = \frac{R_1 \times E_i}{R_1 + \frac{1}{j\omega C}}$$

$$= \frac{E_i}{1 + \frac{1}{jR\omega C}}$$

$$= \frac{E_i}{1 - \frac{j}{\omega RC}}$$

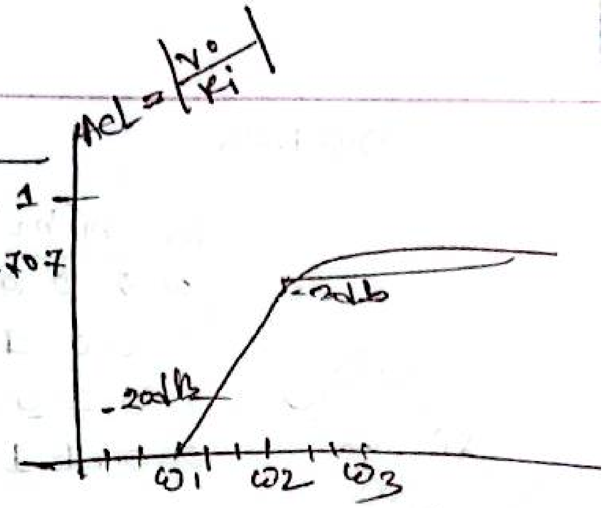
$$V_R = \frac{E}{1 - \frac{j}{\omega RC}}$$

$$V_R = \frac{E_i}{1 - j\left(\frac{1}{\omega RC}\right)}$$

$$\Rightarrow V_1 = R_{R_j} + V_R \quad (1)$$

$$\therefore V_R = \frac{E_i}{1 - j\left(\frac{1}{\omega RC}\right)}$$

$$A_{CL} = \left|\frac{V_o}{R_i}\right| = \frac{1}{1 - j\left(\frac{1}{\omega RC}\right)}$$




$10\log \frac{P_{out}}{P_{in}}$

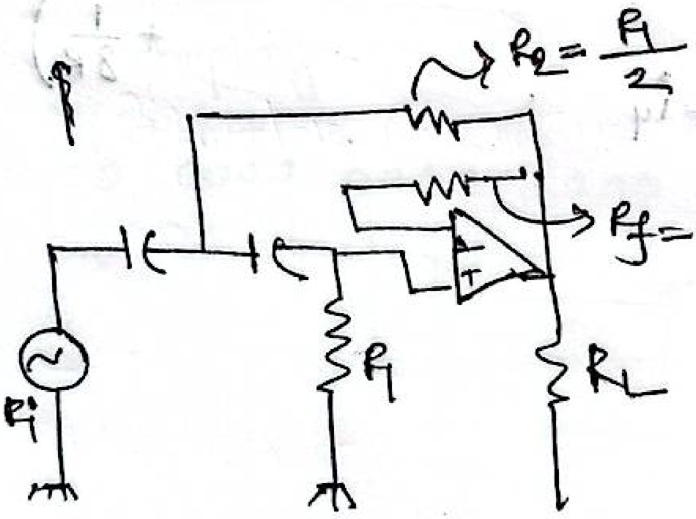
$f_c = 10KHz$

$C = 0.02\ \mu F$

$R = ?$

ω এর স্থান change করলে slope sharp হবে

R এর value বের করে circuit draw করাতে বলে just circuit এ R এর value বসানো লাগবে।



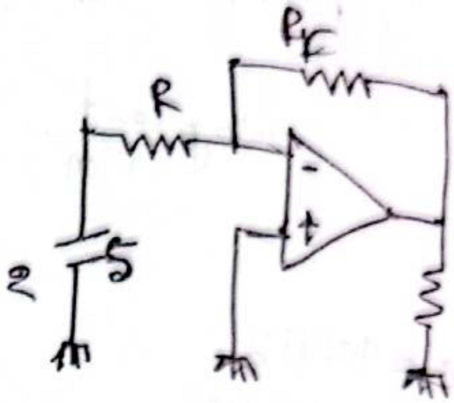

High pass filter with a roll-of -40db diode

19.01.26

Digital – Analog

Analog – Digital

| $b_0$ | $b_1$ | $b_2$ | $b_3$ | | |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | — | 0 |
| 0 | 0 | 0 | 1 | — | 1 |
| 0 | 0 | 1 | 0 | — | 2 |
| 0 | 0 | 1 | 1 | | |




$$V_o = -\frac{R_F}{R} \times 5$$

$$= -\frac{R}{4R} \times 5$$

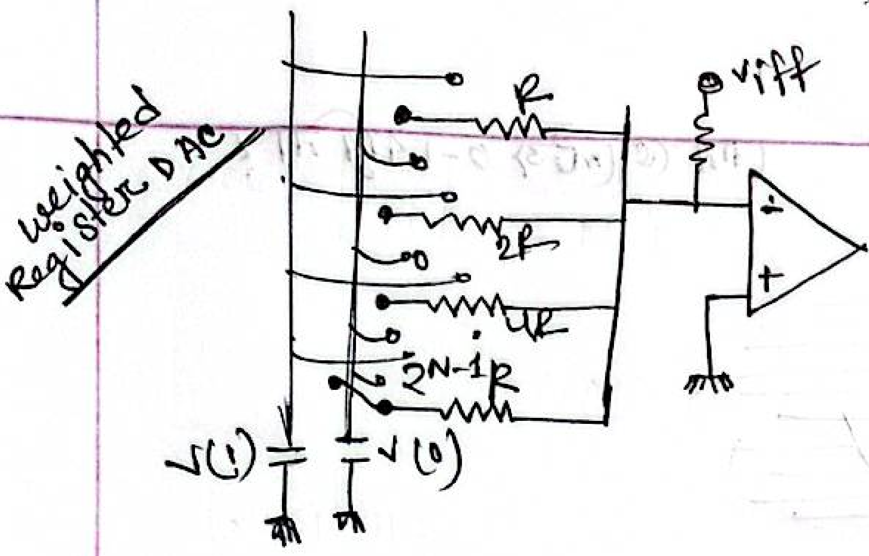
$$= -\frac{5}{4}$$

$$V_o = -\frac{R}{2R} \times 5$$

$$V_o = -V_{in} \times R_F \left(\frac{1}{R} + \frac{1}{2R} + \frac{1}{4R} + \frac{1}{8R}\right)$$

$$i = i_1 + i_2 + i_3 + i_4$$

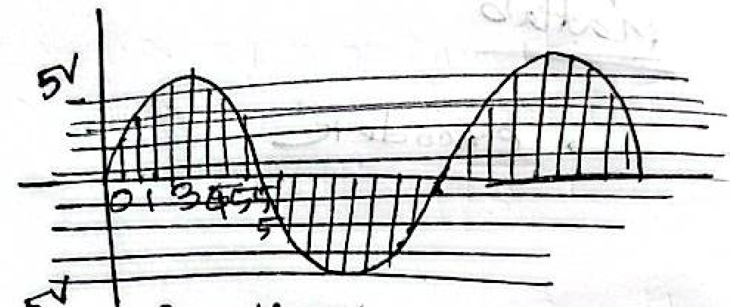
$$i_1 = \frac{V_{in}}{8R}$$




0 0 0
0 0 1
0 1 0
0 1 1
1 0 0
1 0 1
1 1 0
1 1 1.

Fuzzil Logic

*




Sampling

0000 0001 0011 0100

*